



Gosford High School

Year 11

2011

Preliminary Higher School Certificate

Mathematics Extension 1

Assessment Task 3

Time Allowed – 90 minutes

(reading time 5 minutes)

Remember to start each new question in a new booklet

Students must answer questions using a blue/black pen and/or a sharpened B or HB pencil.

Approved scientific calculators may be used

Students need to be aware that

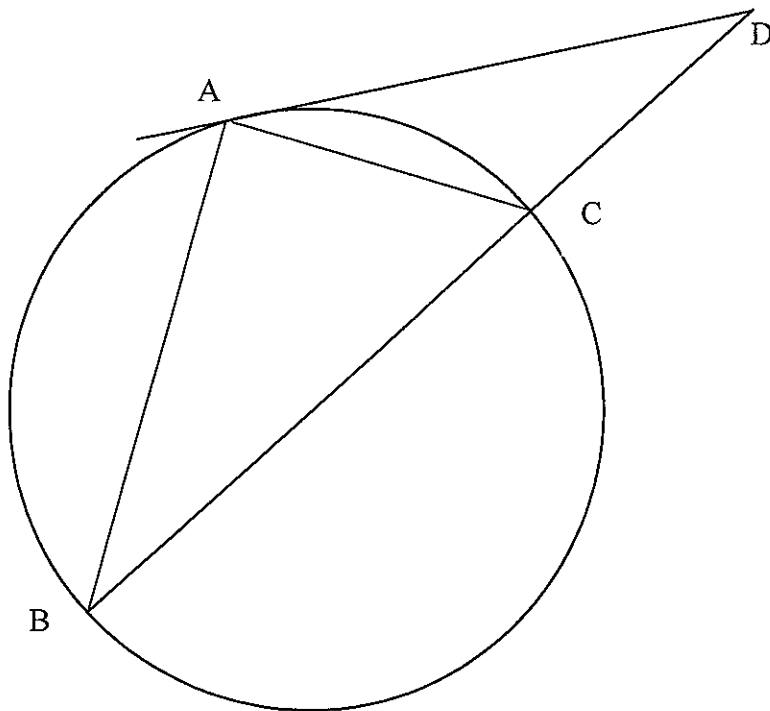
- * 'bald' answers may not gain full marks.
- * untidy and/or poorly organised solutions may not gain full marks.

QUESTION 1 (10 marks) *(start a new booklet)*

- a) For the points $A(-4, 7)$ and $B(-1, 2)$, find the co-ordinates of the point which divides AB externally in the ratio $3 : 1$ (3)

- b) Solve $x - 3 < \frac{4}{x}$ (3)

c)



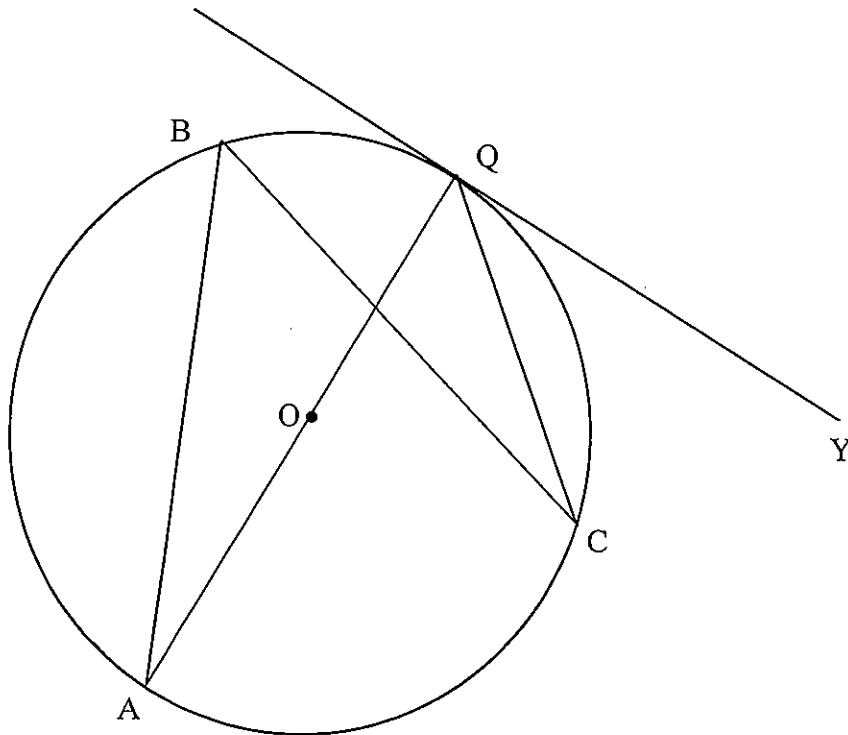
BC is a diameter of a circle. The tangent to the circle at A meets BC produced at D .

$$\angle ABC = 40^\circ.$$

- (i) Find the size of $\angle DAC$, giving reasons (1)
- (ii) Find the size of $\angle CDA$, giving reasons. (3)

QUESTION 2 (10 marks) (*start a new booklet*)

- a) Fully factorise $ap^3 - aq^3 - ap^2q + apq^2$ (2)
- b) Find a linear factor of $4x^3 + 13x^2 + 7x - 6$ (2)
- c) Four couples are to be seated at a round table for a game of Pictionary.
How many different seating arrangements are possible if :
- (i) there are no restrictions? (1)
- (ii) each person sits next to their partner? (2)
- d)



In the diagram QY is a tangent to the circle with centre O.

AQ is a diameter and $\angle YQC = 32^\circ$.

Find $\angle ABC$, giving reasons.

(3)

QUESTION 3 (10 marks) *(start a new booklet)*

a) Find, to the nearest degree, the acute angle formed by the lines $3y = x - 1$ and $4x + 2y - 5 = 0$ (3)

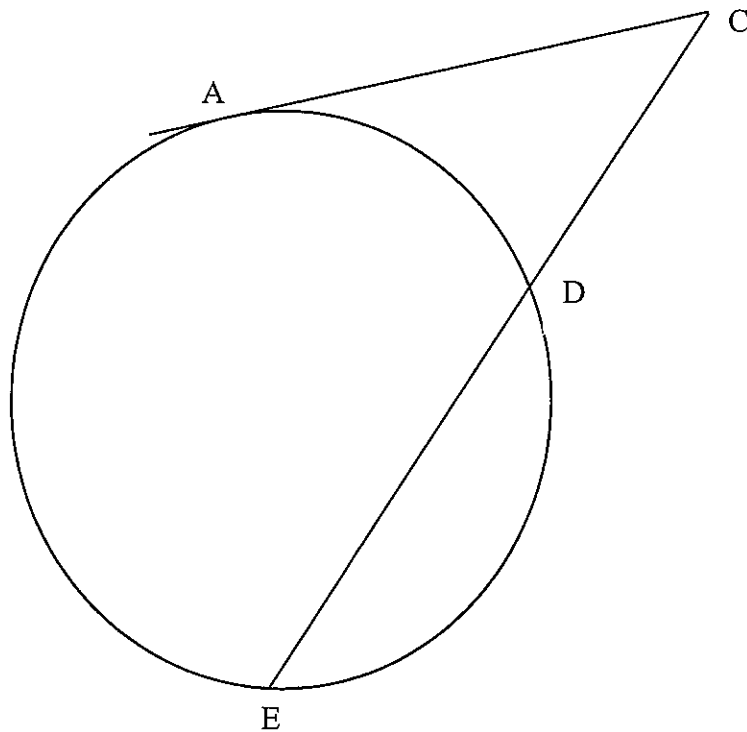
b) Using standard expressions for $\sin 2\theta$ and $\cos 2\theta$, prove that

$$\cos 3\theta = -3\cos\theta + 4\cos^3\theta \quad (4)$$

c) In the diagram the tangent AC intersects the secant EC at C .

Given $AC = 6\text{cm}$ and $ED = 5\text{cm}$.

Find the length of CD , giving reasons. (3)



QUESTION 4 (10 marks) (*start a new booklet*)

- a) A committee of three is to be chosen from a group of four males and five females.

The committee must include at least one male and at least one female.

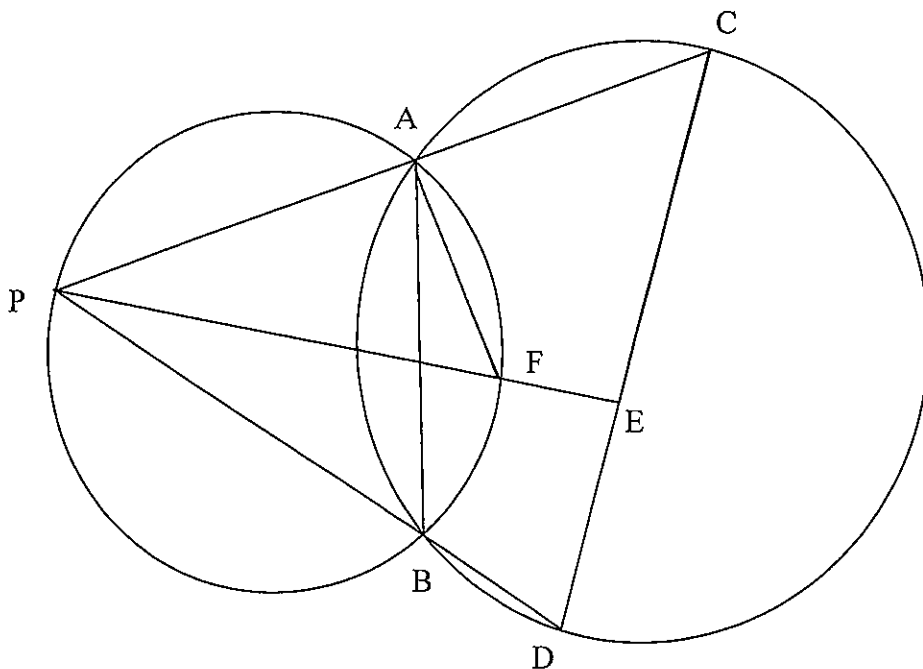
How many different committees can be chosen? (2)

- b) (i) State the sum of the roots of the general quadratic equation $ax^2 + bx + c = 0$ (1)

- (ii) The line $px + y - p^2 = 0$ intersects the parabola $x^2 = 4y$ at A and B .
Show that the x co-ordinates of A and B are the roots of the equation $x^2 + 4px - 4p^2 = 0$. (1)

- (iii) Hence, or otherwise, find the co-ordinates of the midpoint of the interval AB . (2)

c)



In this diagram, two circles intersect at A and B .

P is a point on the first circle such that PA produced and PB produced meet the second circle at C and D respectively.

E is a point on DC such that PE cuts the first circle at F .

- (i) Explain why $\angle PBA = \angle PFA$ (1)

- (ii) Show that $AFEC$ is a cyclic quadrilateral. (3)

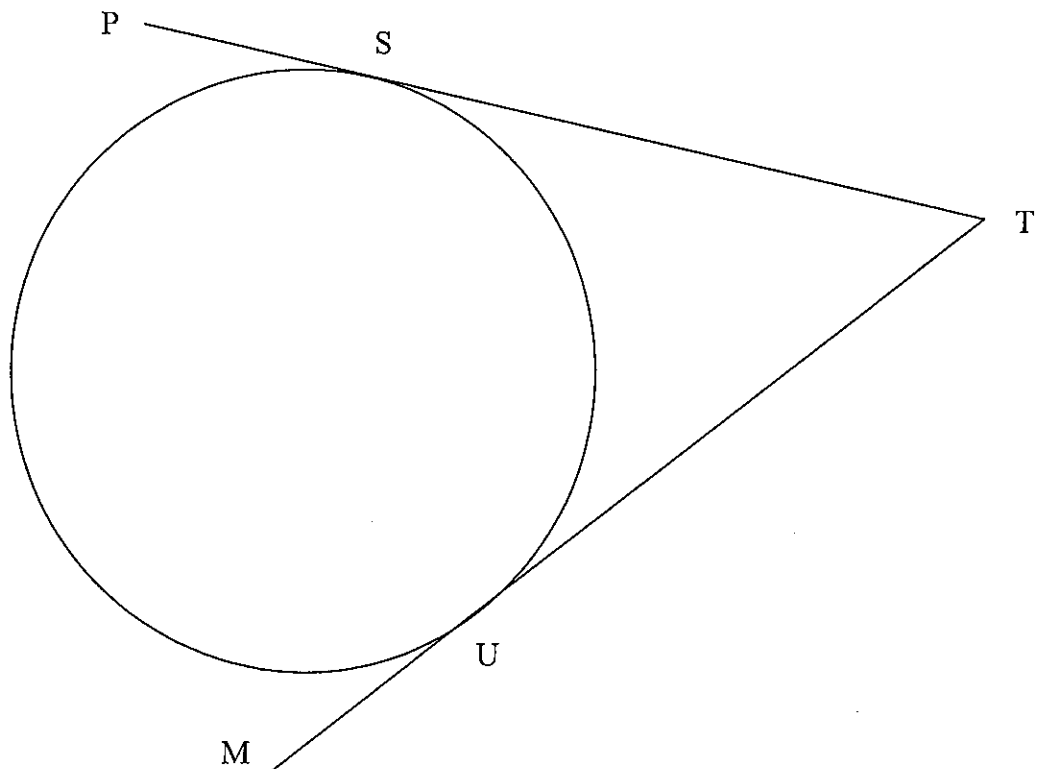
QUESTION 5 (10 marks) (*start a new booklet*)

- a) The polynomial equation $x^3 + bx^2 + cx + d = 0$ has roots α , α^2 and α^3 for some real number $\alpha \neq 0$

(i) Find in terms of c and d the value of $\frac{1}{\alpha} + \frac{1}{\alpha^2} + \frac{1}{\alpha^3}$ (3)

(ii) Show that $b^3d - c^3 = 0$ (2)

b)



S and U are points on a circle. Tangents to the circle at S and U intersect at T .

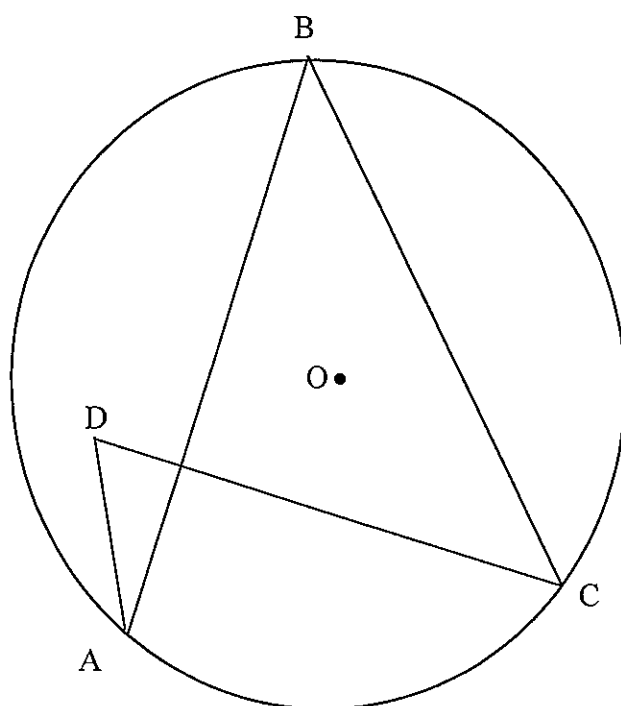
P and M are points on the tangents TS and TU respectively.

R is a point on the circle so that the chord SR is parallel to TU .

(i) Draw a neat diagram showing the given information. (1)

(ii) Prove that $SU = UR$ (2)

c)



Points A , B and C lie on the circumference of a circle with centre O .
The point D lies inside the circle. $\angle ABC = 17^\circ$ and $\angle ADC = 34^\circ$

Prove that $ADOC$ is a cyclic quadrilateral.

(2)

QUESTION 6

(10 marks)

(start a new booklet)

- a) Simplify $\sin 4x \cos 3x - \sin 3x \cos 4x$ (1)
- b) (i) Express $\tan 2x$ in terms of $\tan x$ (1)
- (ii) Hence, or otherwise, show that the exact value of $\tan 15^\circ = 2 - \sqrt{3}$ (3)
- c) In $\triangle ABC$, $\cos B = \frac{1}{3}$ and $\cos C = \frac{2}{3}$.
- (i) Find the exact value of $\sin B$ and $\sin C$ (2)
- (ii) Hence, show that the exact value of $\cos A = \frac{2(\sqrt{10} - 1)}{9}$ (3)

PRELIMINARY HSC EXTENSION 1
MATHEMATICS EXTENSION 1 TASK 3

Question 1

a) Let (x, y) be the required point

$$x = \frac{3(-1) + (-1)(-4)}{2}, \quad y = \frac{3(2) + (-1)(7)}{2}$$

$\therefore$ Point is $(\frac{1}{2}, -\frac{1}{2})$

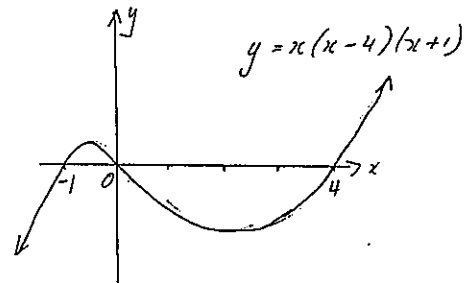
b) $x - 3 < \frac{4}{x}$

$$\therefore x^2(x-3) < 4x$$

$$\therefore x^3 - 3x^2 - 4x < 0$$

$$x(x^2 - 3x - 4) < 0$$

$$x(x-4)(x+1) < 0$$



Using the graph solution is $-1 < x < 0$ and/or $x > 4$

c) (i) $\angle DAC = \angle ABC$ (angle between a tangent and a chord drawn at the point of contact is equal to the angle in the alternate segment.)
 $= 40^\circ$

(ii) $\angle BAC = 90^\circ$ (angle in a semi-circle is a right angle)

$$\begin{aligned} \angle BAD &= \angle BAC + \angle DAC \text{ (adjacent } \angle\text{'s)} \\ &= 40^\circ + 90^\circ \\ &= 130^\circ \end{aligned}$$

$$\begin{aligned} \therefore \angle CDA &= 180^\circ - \angle BAD - \angle ABC \text{ (L. sum of a } \Delta \text{ is } 180^\circ) \\ &= 180^\circ - 130^\circ - 40^\circ \\ &= 10^\circ \end{aligned}$$

Question 2

$$\begin{aligned} \text{a) } ap^3 - aq^3 - ap^2q + apq^2 &= a(p^3 - q^3) - apq(p - q) \\ &= a(p - q)(p^2 + pq + q^2) - apq(p - q) \\ &= a(p - q)(p^2 + pq + q^2 - pq) \\ &= a(p - q)(p^2 + q^2) \end{aligned}$$

$$\begin{aligned} \text{b) } P(-1) &\neq 0, \quad P(-2) = 4(-2)^3 + 13(-2)^2 + 7(-2) - 6 \\ &= -32 + 52 - 14 - 6 \\ &= 0 \end{aligned}$$

$\therefore (x+2)$ is a factor

$$\begin{aligned} \text{c) (i) } \text{N}^\circ. \text{ of arrangements} &= 7! \\ &= 5040 \end{aligned}$$

(ii) Four couples can arrange themselves in $3!$ ways.
Each couple can arrange themselves in 2 ways.
 $\therefore$ N $^\circ$. of arrangements $= 3!(2)^4$
 $= 96$

d) $\angle AXY = 90^\circ$ (L. between a tangent and a chord drawn at the point of contact is a right angle)

$$\begin{aligned} \angle AQC &= 90^\circ - \angle YQC \text{ (adjacent complementary } \angle\text{'s)} \\ &= 90^\circ - 32^\circ \\ &= 58^\circ \end{aligned}$$

$$\begin{aligned} \angle ABC &= \angle AQC \text{ (L's at the circumference standing on the same arc are equal)} \\ &= 58^\circ \end{aligned}$$

Question 3.

a)

$$m_1 = \frac{1}{3}, \quad m_2 = -2$$

Let θ be the required angle

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{\frac{1}{3} + 2}{1 - \frac{2}{3}} \right|$$

$$= \left| \frac{1 + 6}{3 - 2} \right|$$

$$\tan \theta = 7$$

$$\theta = 82^\circ \text{ (to the nearest degree)}$$

b) $\cos 3\theta = \cos(2\theta + \theta)$

$$= \cos 2\theta \cos \theta - \sin 2\theta \sin \theta$$

$$= [2\cos^2 \theta - 1] \cos \theta - 2\sin \theta \cos \theta \cdot \sin \theta$$

$$= 2\cos^3 \theta - \cos \theta - 2\cos \theta (1 - \cos^2 \theta)$$

$$= 2\cos^3 \theta - \cos \theta - 2\cos \theta + 2\cos^3 \theta$$

$$= 4\cos^3 \theta - 3\cos \theta$$

c) $AC^2 = EC \times CD$ (the square of the tangent is equal to the product of the intercepts of the intersecting secant)

Let $x = CD$

$$\therefore 6^2 = (5+x)x$$

$$\therefore 36 = 5x + x^2$$

$$\therefore x^2 + 5x - 36 = 0$$

$$(x+9)(x-4) = 0$$

$$\therefore x = 4 \text{ since } x > 0$$

Question 4

a)

Possible committees have 1 male, 2 females or 2 males, 1 female

$$\begin{aligned} \therefore \text{No. of different committees} &= {}^4C_1 \times {}^5C_2 + {}^4C_2 \times {}^5C_1 \\ &= 4 \times 10 + 6 \times 5 \\ &= 70 \end{aligned}$$

b (i) Sum of roots $= -\frac{b}{a}$

(ii) Solve $y = p^2 - px$ and $x^2 = 4y$ simultaneously

$$\therefore x^2 = 4(p^2 - px)$$

$$\therefore x^2 = 4p^2 - 4px$$

$$\therefore x^2 + 4px - 4p^2 = 0 \text{ is the required equation}$$

(iii) x -coordinate of midpoint $= \frac{\text{Sum of Roots}}{2}$

$$= \frac{-4p}{2}$$

$$= -2p$$

when $x = -2p$, $y = p^2 - p(-2p)$

$$= 3p^2$$

$$\therefore \text{Midpoint is } (-2p, 3p^2)$$

Question 4 (continued)

- (ii) $\angle PBA = \angle PFA$ ($\angle$'s. at the circumference standing on the same arc (or chord) are equal)

(ii) Let $\angle PBA = \angle PFA = x$

$\therefore \angle ABD = 180^\circ - x$ (adjacent supplementary $\angle$'s)

$$\begin{aligned} \therefore \angle ACE &= 180^\circ - (180^\circ - x)^\circ \quad (\text{the opposite } \angle\text{'s. of} \\ &= x \quad \text{cyclic quadrilateral ABDC} \\ &= \angle AFP \quad \text{are supplementary}) \end{aligned}$$

$\therefore$ AFEC is a cyclic quadrilateral as one angle of the quadrilateral AFEC is equal to the opposite exterior angle.

Question 5

a) (i) Sum of roots = $-\frac{b}{a}$.

$$\therefore \alpha + \alpha^2 + \alpha^3 = -6$$

Sum of roots
taken two at
a time $= \frac{c}{a}$.

$$\therefore \alpha^3 + \alpha^4 + \alpha^5 = C$$

and Product of Roots = $-\frac{d}{a}$.

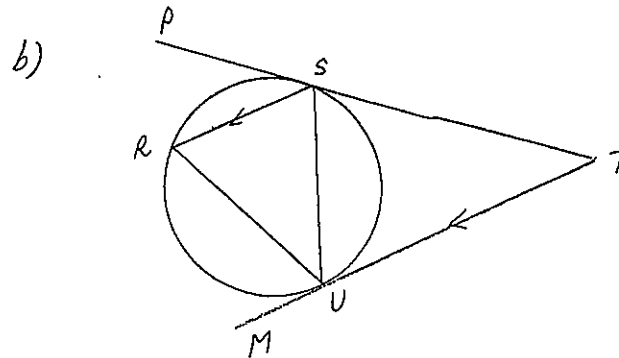
$$\therefore \alpha^6 = -d$$

$$\text{Now } \frac{1}{a} + \frac{1}{a^2} + \frac{1}{a^3} = \frac{a^3 + a^2 + a}{a^6} = \frac{c}{-d}$$

$$\begin{aligned} \text{(ii)} \quad C &= \alpha^3 + \alpha^4 + \alpha^5 \\ &= \alpha^2 (\alpha + \alpha^2 + \alpha^3) \\ &= \alpha^2 \cdot (-b) \\ C^3 &= \alpha^6, -b^3. \end{aligned}$$

$$\therefore c^3 = -d \cdot (-b^3).$$

$$b^3 d - c^3 = 0$$



Question 5 (continued)

$$\therefore \angle MOR = \angle SRU \quad (\text{alternate } \angle's, MU \parallel RS)$$

$$\angle MUR = \angle RSU \quad (\angle \text{ between a tangent and a chord is equal to the angle in the alternate segment})$$

$$\therefore \angle SRU = \angle RSU (= \angle MUR)$$

$$\therefore SU = UR \quad (\text{equal sides opposite equal } \angle's \text{ of isosceles } \Delta)$$

$$\begin{aligned} \text{c) } \angle AOC &= 2 \times \angle ABC \quad (\text{angle at the centre is twice the angle at the circumference, standing on the same arc}) \\ &= 2 \times 17^\circ \\ &= 34^\circ \end{aligned}$$

$$\therefore \angle AOC = \angle ADC (= 34^\circ)$$

$\therefore ADOC$ is a cyclic quadrilateral

(interval AC subtends two equal $\angle$'s at D $\neq$ O on the same side of it.)

Question 6

$$\begin{aligned} \text{a) } \sin 4x \cos 3x - \sin 3x \cos 4x &= \sin(4x - 3x) \\ &= \sin x \end{aligned}$$

$$\text{b) (i) } \tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$

$$\text{(ii) } \tan 30^\circ = \frac{2 \tan 15^\circ}{1 - \tan^2 15^\circ}$$

$$\frac{1}{\sqrt{3}} = \frac{2 \tan 15^\circ}{1 - \tan^2 15^\circ}$$

$$1 - \tan^2 15^\circ = 2\sqrt{3} \tan 15^\circ$$

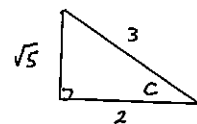
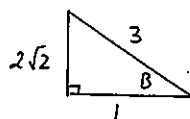
$$\therefore \tan^2 15^\circ + 2\sqrt{3} \tan 15^\circ - 1 = 0$$

$$\therefore \tan 15^\circ = \frac{-2\sqrt{3} \pm \sqrt{12 + 4}}{2}$$

$$\tan 15^\circ = \frac{-2\sqrt{3} \pm 4}{2}$$

$$\therefore \tan 15^\circ = -2\sqrt{3} + 2 \quad \text{since } \tan 15^\circ > 0$$

(*)



$$\text{(i) } \sin B = \frac{2\sqrt{2}}{3}$$

$$\sin C = \frac{\sqrt{5}}{3}$$

$$\begin{aligned} \text{(ii) } \cos A &= \cos(180 - (B + C)) \quad ABC \text{ is a } \Delta \\ &= -\cos(B + C) \end{aligned}$$

$$= -[\cos B \cos C - \sin B \sin C]$$

$$= -\left[\frac{2\sqrt{2}}{3} \cdot \frac{\sqrt{5}}{3} - \frac{1}{3} \cdot \frac{2}{3} \right]$$

$$\therefore \cos A = \frac{2\sqrt{10} - 2}{9} = \frac{2(\sqrt{10} - 1)}{9}$$



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Preliminary

Higher School Certificate

Mathematics Extension 1

Assessment Task 3

Paper 1

Time Allowed – 50 minutes

Remember to start each new question on a new page

Students must answer questions using a blue/black pen and/or a sharpened B or HB pencil.

Approved scientific calculators may be used

Students need to be aware that

- * 'bald' answers may not gain full marks.
- * untidy and/or poorly organised solutions may not gain full marks.

POLYNOMIALS Part A (start a new page)

- 1) Sketch the graph of $y = 2x^2(1-x)^3$ (no calculus required) (2)
- 2) Find the remainder when $x^3 - x^2 + 11x - 17$ is divided by $x + 1$ (2)
- 3) Given $P(x) = x^4 - 3x^3 - 2x + 7$ and $A(x) = x^2 + 3$,
- (i) find the quotient $Q(x)$ and the remainder $R(x)$ when $P(x)$ is divided by $A(x)$. (2)
- (ii) Express your result in the form $P(x) = A(x) \cdot Q(x) + R(x)$. (1)
- 4) When $P(x)$ is divided by $x^2 - 3x - 4$ the remainder is $2x + 3$.
Find the remainder when $P(x)$ is divided by $x - 4$. (2)
- 5) If α, β and γ are the roots of the equation $2x^3 - 4x^2 + 5x - 3 = 0$, find the values of
- (i) $\alpha + \beta + \gamma$ (ii) $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$ (iii) $\alpha^2 + \beta^2 + \gamma^2$ (5)

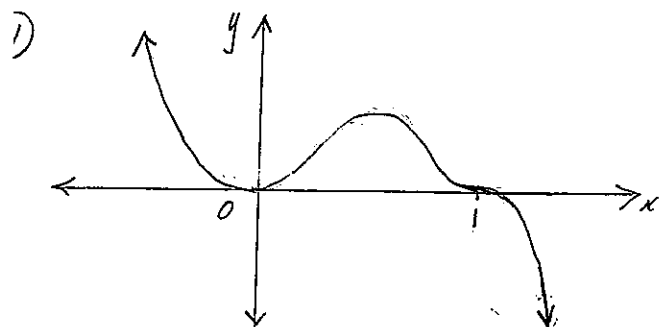
POLYNOMIALS Part B (start a new page)

- 6) (i) If $P(x) = 2x^3 + x^2 - 13x + 6$, find the value of $P(2)$ (1)
- (ii) Hence, solve $2x^3 + x^2 - 13x + 6 = 0$ (3)
- 7) The polynomials x^3 and $A(x-1)(x-2)(x-3) + B(x-1)(x-2) + C(x-1) + D$ are equal for more than 3 values of x . Find the values of the constants A, B, C and D. (4)
- 8) Two of the roots of $2x^4 - 5x^3 + cx^2 + 45x - 18 = 0$ are reciprocals, while the remaining two roots are opposites of each other. Find the four roots and hence evaluate c . (4)

INEQUALITIES (start a new page)

- 1) Solve $2x(x-4)(3-x) > 0$ (2)
- 2) Solve $\frac{x^2 - 5x}{x - 4} \leq 3$ (3)
- 3) Find the set of values of x for which $x^2 < 25$ and $\frac{5}{|x-4|} \leq 1$ hold simultaneously. (3)

Polynomials Part A.



② Let $P(x) = x^3 - x^2 + 11x - 17$
Remainder = $P(-1)$
$$= (-1)^3 - (-1)^2 + 11(-1) - 17$$
$$= -1 - 1 - 11 - 17$$
$$= -30$$

③ (i)

$$\begin{array}{r} x^2 + 3 \overline{) x^4 - 3x^3 - 2x + 7} \\ \underline{x^4 + 3x^2} \\ -3x^3 - 3x^2 - 2x + 7 \\ \underline{-3x^3 - 9x} \\ -3x^2 + 7x + 7 \\ \underline{-3x^2 - 9} \\ 7x + 16 \end{array}$$

(ii)
 $\therefore P(x) = (x^2 + 3)(x^2 - 3x - 3) + 7x + 16$

④ $P(x) = (x^2 - 3x - 4) \cdot A(x) + (2x + 3)$
$$= (x - 4)(x + 1) \cdot A(x) + (2x + 3)$$

Remainder when divided
by $(x - 4)$ is $P(4)$

$$P(4) = 0 + 2(4) + 3$$
$$= 11$$

Remainder is 11

⑤ (i) $\alpha + \beta + \gamma = \frac{-b}{a}$
$$= 2$$

(ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$
$$= \frac{5/2}{3/2}$$
$$= \frac{5}{3}$$

(iii) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - \sum 2\alpha\beta$
$$= 2^2 - 2 \times \frac{3}{2}$$
$$= 1$$

Polynomials Part B.

⑥ (i) $P(2) = 2(2)^3 + (2)^2 - 13(2) + 6$
$$= 16 + 4 - 26 + 6$$
$$= 0$$

$\therefore x - 2$ is a factor

$$\begin{array}{r} 2x^2 + 5x - 3 \\ x - 2 \overline{) 2x^3 + x^2 - 13x + 6} \\ \underline{2x^3 - 4x^2} \\ 5x^2 - 13x + 6 \\ \underline{5x^2 - 10x} \\ -3x + 6 \\ \underline{-3x + 6} \\ 0 \end{array}$$

$\therefore (x - 2)(2x^2 + 5x - 3) = 0$
 $(x - 2)(2x - 1)(x + 3) = 0$
 $\therefore x = 2, \frac{1}{2}, -3$

$$7) x^3 \equiv A(x-1)(x-2)(x-3) + B(x-1)(x-2) + C(x-1) + D$$

True for $x=1$ $\therefore 1 = D$.

True for $x=2$ $\therefore 8 = C + 1$
 $7 = C$

True for $x=3$ $\therefore 27 = B(2)(1) + 7(2) + 1$
 $12 = 2B$
 $B = 6$

and $A=1$ by inspection of coefficient of x^3 .
 $\therefore A=1, B=6, C=7, D=1$

⑧ Let the roots be
 $\alpha, \frac{1}{\alpha}, \beta, -\beta$

$$\sum \alpha = -\frac{b}{a}$$

$$\therefore \alpha + \frac{1}{\alpha} = \frac{5}{2}$$

$$2\alpha^2 + 2 = 5\alpha$$

$$\therefore 2\alpha^2 - 5\alpha + 2 = 0$$

$$(2\alpha - 1)(\alpha - 2) = 0$$

$$\therefore \alpha = \frac{1}{2}, 2$$

Product of Roots = $\frac{e}{a}$

$$-\beta^2 = -9$$

$$\beta^2 = 9$$

$$\beta = \pm 3$$

$\therefore$ Roots are $2, \frac{1}{2}, 3, -3$

$x=2$ satisfies

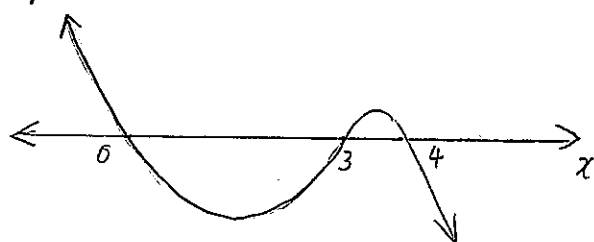
$$\therefore 32 - 40 + 4C + 90 - 18 = 0$$

$$4C = -64$$

$$C = -16$$

Inequalities

①



$$\{x : x < 0\} \cup \{x : 3 < x < 4\}$$

② Critical values $x \neq 4$

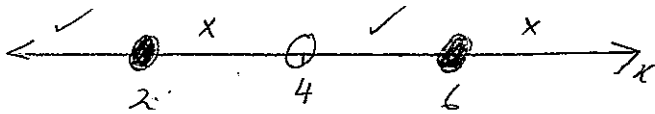
and when $\frac{x^2 - 5x}{x - 4} = 3$

$$x^2 - 5x = 3x - 12$$

$$x^2 - 8x + 12 = 0$$

$$(x-6)(x-2) = 0$$

$$x = 2, 6$$



Test $x = 0$, True

Test $x = 3$, False

Test $x = 5$, True

Test $x = 7$, False

$$\therefore \{x : x \leq 2\} \cup \{x : 4 < x \leq 6\}$$

$$③ \quad x^2 < 25$$

$$\therefore -5 < x < 5$$

$$5 \leq |x-4|$$

$$\therefore |x-4| \geq 5$$

$$\therefore x-4 \leq -5 \text{ or } x-4 \geq 5$$

$$x \leq -1 \quad x \geq 9$$

$\therefore$ Require set of points satisfying

$$-5 < x < 5 \text{ and}$$

$$x \leq -1 \cup x \geq 9$$

$$\therefore \underline{\underline{-5 < x \leq -1}}$$



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Mathematics Extension 1

Assessment Task 3 Paper 2

Time Allowed – 60 minutes

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COUNTING TECHNIQUES *(start a new page)*

- 1) In how many ways can the letters of the word CYCLIC be arranged
 - (i) without any restrictions? (1)
 - (ii) if all the C's are to be together? (2)
 - (iii) if there is to be a vowel on an end and *no* C's are to be together? (2)

- 2) In how many ways can 4 boys and 4 girls be seated around a table if
 - (i) there are no restrictions (1)
 - (ii) the boys and girls alternate (1)
 - (iii) two particular boys wish to sit next to each other (1)
 - (iv) two particular girls wish to sit opposite each other. (1)

- 3) A committee of 5 is to be chosen from 6 boys and 4 girls.
 - (a) Find the total number of different committees possible (1)
 - (b) Find the total number of different committees possible if the committee contains
 - (i) 3 boys and 2 girls. (1)
 - (ii) a particular boy X but excludes a particular girl Y. (1)
 - (iii) a majority of girls (2)

DIVISION OF A LINE IN A GIVEN RATIO *(start a new page)*

- 4) A (2, -1) and B (-5, -3) are two points.
Find the coordinates of the point S which divides AB externally in the ratio 4 : 7. (3)
- 5) The points A (-1, 5), B (6, -4) and P (20, -22) are collinear.
Find the ratio in which P divides the interval AB. (3)
- 6) P (x, y) divides the join of (-1, 0) and (3, 5) in the ratio $k : 1$.
Find the values of k for which $x + y > 4$. (4)

HARDER 2 UNIT *(start a new page)*

- 7) The length of a rectangle is increased by 12% and the width is decreased by 10%.
What is the percentage change in the area. (1)
- 8) Find x and y given that $25^x = 125^y$ and $\frac{16^x}{8} = 2 \times 4^{2y}$ (3)
- 9) Make x the subject of the equation $t = \frac{1}{2(4-x)} - \frac{1}{8}$, expressing your answer as a single fraction. (3)
- 10) Factorise $m^3 - a^3 - m^2 + 2am - a^2$ (3)
- 11) Solve $x^2 - 4x + \frac{5}{x^2 - 4x} = 6$ (3)
- 12) Find the average of the roots of $2x^2 + 14x + 17 = 0$ (2)

Task 3 (Paper 2)Counting Techniques

① (i) N° of arrangements = $\frac{6!}{3!} = 120$ ways

(ii) c's can only arrange themselves as a group in 1 way
then the c's (as one) and the other 3 letters
can arrange themselves in $4!$ ways

$\therefore N^{\circ}$ of arrangements = $1 \times 4! = 24$ ways

(iii) $\boxed{i} \boxed{c} \boxed{2} \boxed{c} \boxed{1} \boxed{c}$

with vowel (i) in first position c's must occupy
alternate positions resulting in $2!$ ways.

But vowel (i) can be on the end with c's again
in alternate positions resulting again in $2!$ ways

$\therefore N^{\circ}$ of arrangements = $2 \times 2! \text{ ways}$
= 4 ways

② i) N° of arrangements = $(8-1)! = 7! = 5040$ ways

ii) N° of arrangements = $3! 4! = 144$ ways

iii) N° of arrangements = $2 \times 6! = 1440$ ways

iv) N° of arrangements = $1 \times 6! = 720$ ways.

$$(3) \quad a) \quad N^{\circ} \text{ of Committees} = {}^{10}C_5 = 252$$

$$b) \quad i) \quad N^{\circ} \text{ of Committees} = {}^6C_3 \times {}^4C_2 = 120$$

$$ii) \quad N^{\circ} \text{ of Committees} = {}^8C_4 = 70$$

$$iii) \quad N^{\circ} \text{ of Committees} = {}^4C_4 \times {}^6C_1 + {}^4C_3 \times {}^6C_2$$

$$= 6 + 60$$

$$= 66$$

Division of a Line in a Given Ratio

$$(4) \quad A(2, -1) \quad B(-5, -3) \quad S(x, y)$$

$$\begin{array}{c} \nearrow \searrow \\ -4 : 7 \end{array}$$

$$\therefore x = \frac{(-4)(-5) + 7(2)}{7-4}$$

$$x = \frac{34}{3}$$

$$\text{and } y = \frac{(-4)(-3) + 7(-1)}{3}$$

$$= \frac{5}{3}$$

$$\therefore S\left(11\frac{1}{3}, 1\frac{2}{3}\right)$$

$$(5) \quad \text{Let the ratio be } k:1$$

$$A(-1, 5) \quad B(6, -4)$$

$$\begin{array}{c} \nearrow \searrow \\ k : 1 \end{array}$$

$$\therefore \frac{6k-1}{k+1} = 20$$

$$6k-1 = 20k+20$$

$$-21 = 14k$$

$$-\frac{3}{2} = k$$

$$\therefore \text{Ratio is } -\frac{3}{2} : 1$$

$$\text{i.e. } -3 : 2 \quad \text{or } 3 : -2 \quad \text{or } 3 : 2 \quad \text{Externally}$$

⑥ $(-1, 0) \xleftrightarrow{k:1} (3, 5)$

$$x = \frac{3k-1}{k+1} \quad \text{and} \quad y = \frac{5k}{k+1}$$

$$\therefore \frac{3k-1}{k+1} + \frac{5k}{k+1} > 4$$

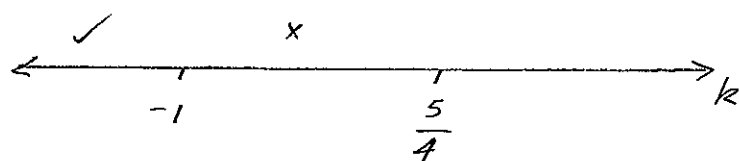
$$\therefore \frac{8k-1}{k+1} > 4$$

Critical values occur when $k=1 \neq$ when $\frac{8k-1}{k+1} = 4$

$$\text{ie } 8k-1 = 4k+4$$

$$4k = 5$$

$$k = \frac{5}{4}$$



Test $k < -1$ (say $k = -2$) True.

$-1 < k < \frac{5}{4}$ (say $k = 0$) False

$k > \frac{5}{4}$ (say $k = 2$) True

$\therefore$ Solution is $k < -1$ or $k > \frac{5}{4}$.

Harder 2 Unit

⑦ Let original dimensions be x, y $\therefore$ Area = xy .

New dimensions are $1.12x, 0.9y$ $\therefore$ Area = $1.008xy$.

$\therefore$ % change is 0.8%

⑧

$$5^{2x} = 5^{3y}$$

$$\therefore 2x = 3y \dots (1)$$

$$\frac{2^{4x}}{2^3} = 2 \times 2^{4y}$$

$$2^{4x-3} = 2^{4y+1}$$

$$\therefore 4x-3 = 4y+1$$

$$\text{But } 4x = 6y \text{ from (1)}$$

$$6y-3 = 4y+1$$

$$2y = 4$$

$$y = 2 \text{ and } \therefore x = 3$$

⑨

$$t + \frac{1}{8} = \frac{1}{2(4-x)}$$

$$8t+1 = \frac{4}{4-x}$$

$$4-x = \frac{4}{8t+1}$$

$$x = 4 - \frac{4}{8t+1}$$

$$x = \frac{4(8t+1)-4}{8t+1}$$

$$x = \frac{32t}{8t+1}$$

$$(10) (m-a)(m^2+am+a^2) - (m^2-2am+a^2)$$

$$= (m-a)(m^2+am+a^2) - (m-a)^2$$

$$= (m-a)(m^2+am+a^2-m+a)$$

⑪ Let $u = x^2 - 4x$

$$\therefore u + \frac{5}{u} = 6$$

$$u^2 + 5 = 6u$$

$$u^2 - 6u + 5 = 0$$

$$(u-5)(u-1) = 0$$

$$u = 5, 1$$

If $u = 5$ then $x^2 - 4x = 5$

$$x^2 - 4x - 5 = 0$$

$$(x-5)(x+1) = 0$$

$$x = 5, -1$$

If $u = 1$ then $x^2 - 4x = 1$

$$x^2 - 4x - 1 = 0$$

$$x = \frac{4 \pm \sqrt{16+4}}{2}$$

$$x = \frac{4 \pm \sqrt{20}}{2}$$

$$x = 2 \pm \sqrt{5}$$

$$\therefore x = -1, 5, 2 \pm \sqrt{5}$$

⑫ Sum of Roots = $-\frac{b}{a}$

$$= -7$$

Average of Roots = $\frac{\text{Sum of Roots}}{2}$

$$= \frac{-7}{2}$$